

Project Title

AI-Based Smart Transport Fleet Management System

SOA Project Review-1

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Under the Guidance of

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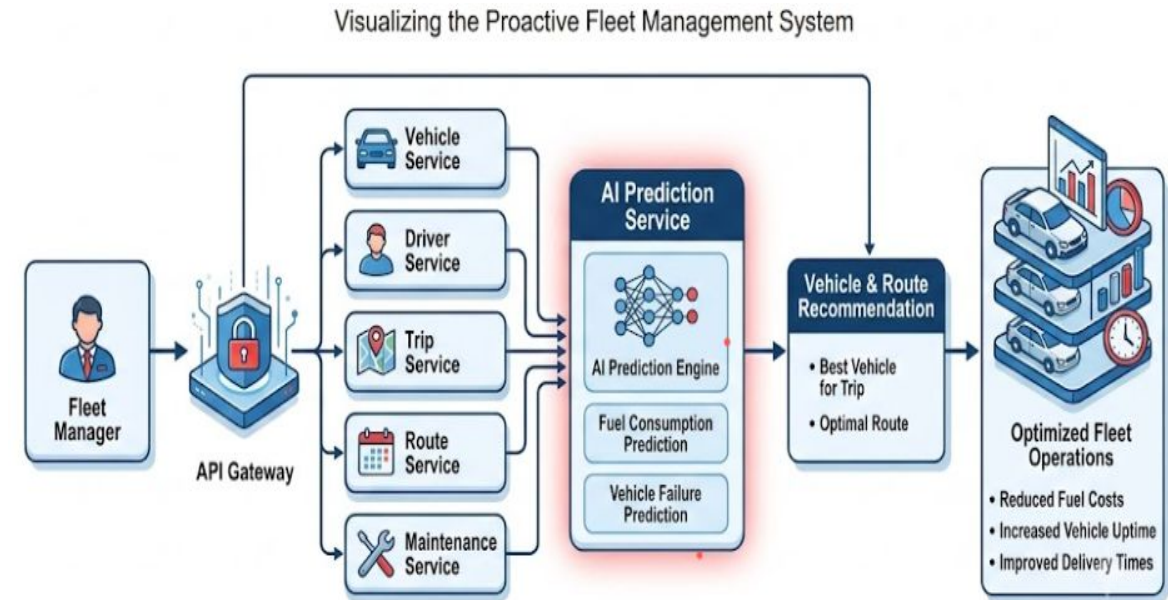
SOA Project Review-1

Outline of Project Work



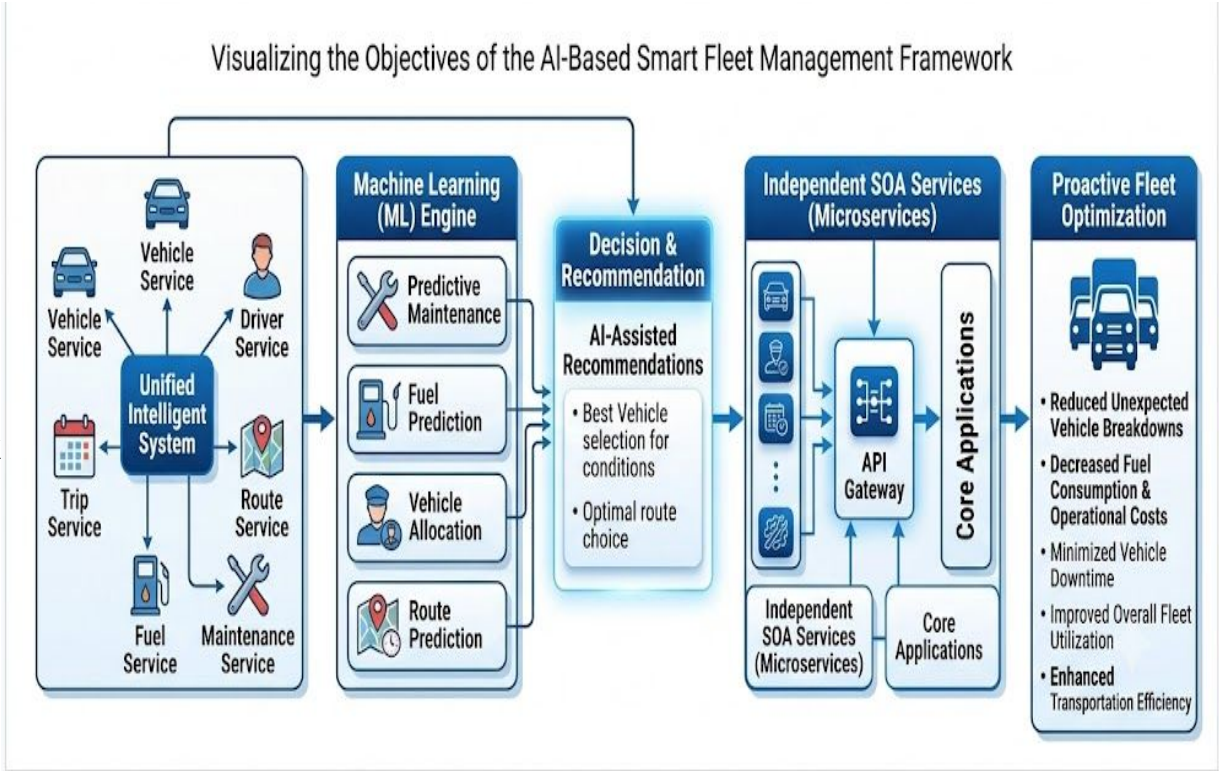
Problem Statement

- Traditional fleet management systems mainly depend on **manual decision-making and reactive management**.
- **Unexpected vehicle breakdowns** can lead to vehicle downtime and increased maintenance costs.
- Inefficient **vehicle allocation** can result in poor fleet utilization and underutilization of available vehicles.
- High and unpredictable **fuel consumption** can increase overall transportation and operational costs.
- Inefficient **route and trip planning** can result in longer travel times and unnecessary fuel usage.
- Existing systems often manage **vehicles, drivers, trips, fuel, and maintenance separately**, limiting integrated decision-making.
- There is a need for an **AI-powered SOA-based system** that predicts risks and recommends efficient fleet decisions proactively.



Objectives

- To improve** fleet management efficiency by integrating vehicle, driver, trip, route, fuel, and maintenance information through a unified intelligent system.
- To leverage** Machine Learning techniques for predictive maintenance, fuel-consumption prediction, vehicle allocation, and travel-time/route prediction.
- To enhance the existing system, we propose an AI-Based Smart Fleet Management Framework** that integrates independent SOA-based services for intelligent and scalable fleet operations.
- To provide** AI-assisted recommendations for selecting the most suitable vehicle and route based on vehicle condition, availability, fuel efficiency, maintenance risk and trip requirements.
- To develop** an SOA-based modular architecture in which vehicle, driver, trip, maintenance, route, and AI services communicate through APIs and can be independently developed, deployed, and scaled.
- To reduce** unexpected vehicle breakdowns, fuel consumption, operational costs, and vehicle downtime through predictive and data-driven decision-making.
- To improve** overall fleet utilization and transportation efficiency by enabling proactive planning and intelligent fleet optimization.



Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Theissler et al. (2021)	Predictive maintenance enabled by machine learning: Use cases and challenges in the automotive industry	ML-based predictive maintenance; supervised and deep-learning approaches	ML can use vehicle operating data to predict maintenance needs and support vehicle reliability.	Requires sufficient labelled data; combining multiple data sources and improving interpretability remain challenges.
Monnerat et al. (2019)	Fleet management: A vehicle and driver assignment model	Generalized Assignment Problem; mathematical optimization and genetic-algorithm-based matheuristic	Joint vehicle-driver assignment can optimize planned trips and reduce operational costs.	Larger fleet-assignment problems become computationally challenging; heuristic approaches are needed for scalability.
Ko (2022)	Machine Learning Methods to Predict Vehicle Fuel Consumption	Linear Regression, KNN, Decision Tree, Random Forest, Gradient Boosting	Vehicle speed, acceleration and road gradient can be used as features for ML-based fuel-consumption prediction.	Prediction depends on driving and road conditions; broader real-world datasets can improve applicability.
Giannoulidis & Gounaris (2023)	A context-aware unsupervised predictive maintenance solution for fleet management	Unsupervised streaming anomaly detection; proximity-based, statistical and transformer methods	Context-aware anomaly detection can support predictive maintenance in vehicle fleet environments.	Detecting failures in advance remains challenging; wider data availability and real-world validation can improve robustness.

Problem Gap Identification

Problem Gap Identification

Existing approaches

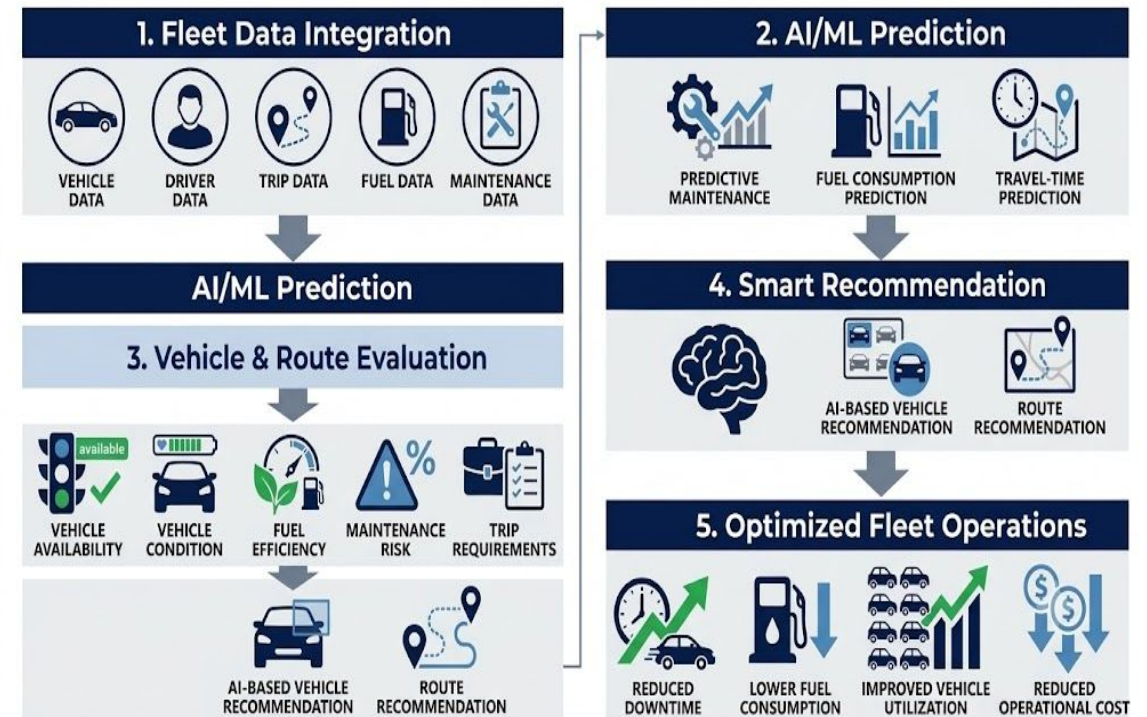
- Primarily focus on **vehicle tracking, trip scheduling, and record management**.
- Fleet decisions often depend on **manual and reactive decision-making**.
- Maintenance is generally performed **after faults or breakdowns occur**.
- Vehicle allocation may not jointly consider **availability, condition, fuel efficiency, and maintenance risk**.
- Fuel monitoring and route planning are often handled **independently**.
- Monolithic systems can make **scaling, integration, and maintenance** difficult.

Identified Gap

There is a need for an integrated intelligent fleet-management system that combines:

AI/ML Prediction + Predictive Maintenance + Fuel Prediction + Vehicle Allocation + Route/Travel-Time Prediction + SOA Architecture

Our Approach: AI-Based Fleet Optimization



Innovation, Creativity, and Novelty

1. AI-Based Vehicle Allocation

The system recommends the most suitable vehicle for each trip by considering:

- Vehicle availability
- Vehicle condition
- Fuel efficiency
- Maintenance risk
- Trip requirements

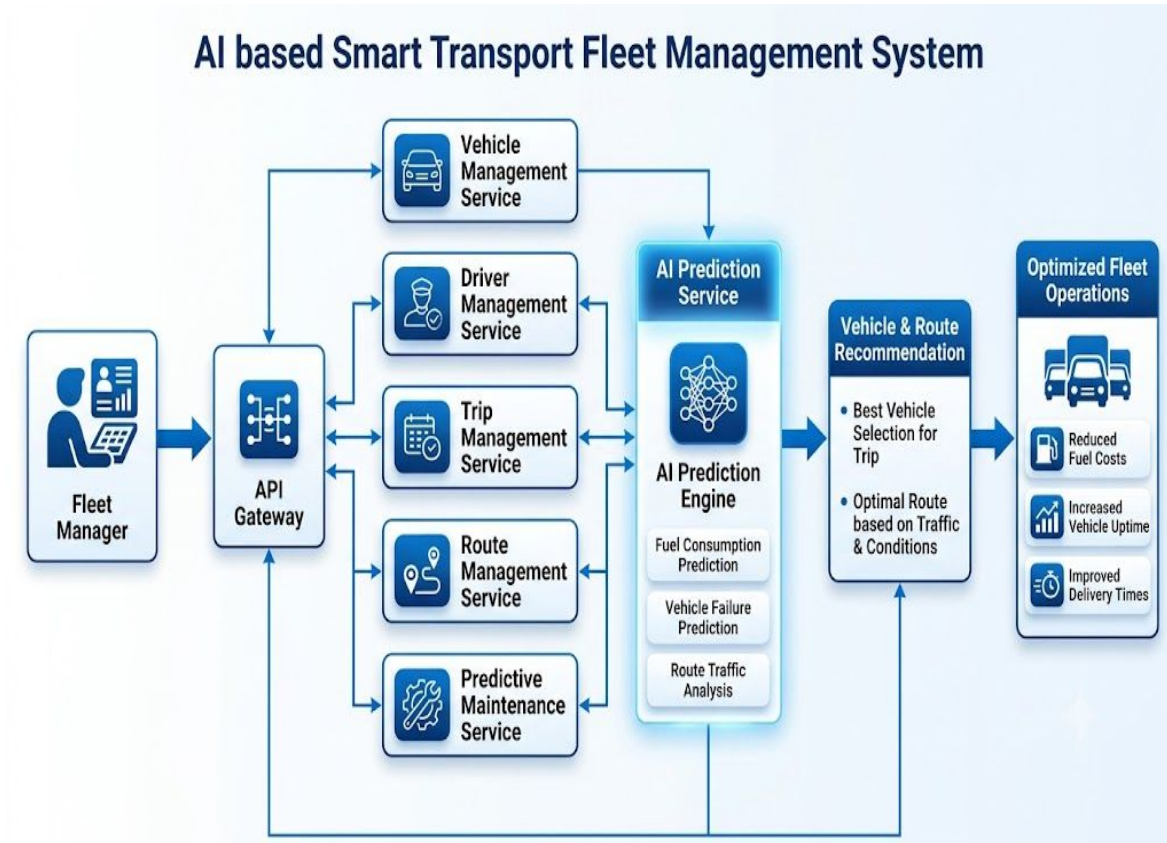
2. Predictive Fleet Intelligence

Machine Learning predicts important fleet parameters such as:

- Maintenance requirements
- Fuel consumption
- Travel time
- Route efficiency

3. Integrated SOA-Based Fleet Optimization

Instead of managing fleet operations separately, the system integrates **vehicle, driver, trip, route, maintenance, and AI services** through APIs to provide intelligent and scalable fleet-management decisions.



4. Intelligent Route & Travel-Time Prediction

Uses historical and operational data to predict travel time and recommend efficient routes for planned trips.

5. Integrated Fleet Decision-Making

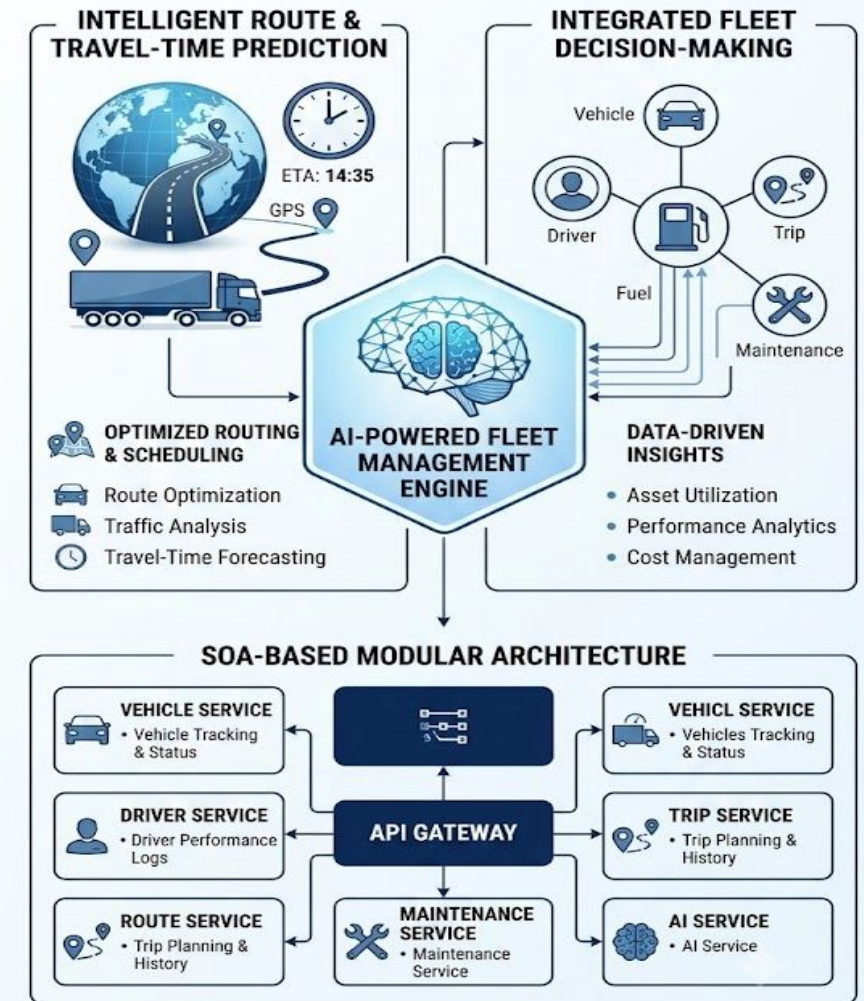
Combines vehicle, driver, trip, maintenance, fuel, and route information to provide data-driven fleet management decisions.

6. SOA-Based Modular Architecture

Implements independent services communicating through APIs, allowing vehicle, driver, trip, route, maintenance, and AI services to be developed, deployed, and scaled independently.

Integrated AI Fleet Intelligence

AI-Powered Fleet Management System



Feasibility and Project Plan

✓ FEASIBILITY

- **Technical Feasibility:** The system uses established technologies including React.js, Spring Boot, Python FastAPI, Scikit-learn, PostgreSQL, Spring Cloud, and REST APIs, supporting scalable service-based development.
- **Operational Feasibility:** The system provides separate services for vehicles, drivers, trips, routes, maintenance, fuel, location, notifications, users, and analytics, enabling efficient fleet operations through a unified platform.
- **Economic Feasibility:** The proposed system primarily uses open-source technologies and modular SOA services, reducing development and maintenance costs while allowing individual services to scale independently.
- **Schedule Feasibility:** The project is divided into 10 phases, covering requirement analysis, SOA service development, AI/ML integration, database integration, testing, and final deployment within the planned timeline.

PROJECT TIMELINE (10 WEEKS)

Phase	Work / Deliverables	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
1	Requirement Analysis & Research										
2	Literature Survey & Dataset Preparation										
3	SOA Architecture & API Gateway Design										
4	Vehicle, Driver & Trip Services Development										
5	Route, Location, Fuel & Maintenance Services										
6	AI/ML Model Development & Prediction										
7	Vehicle Allocation & Fleet Recommendation										
8	Database & External API Integration										
9	Frontend, Integration & Performance Testing										
10	Security, Deployment & Final Evaluation										



Spring Boot
Microservices



Spring Cloud
(Eureka, Gateway)



PostgreSQL
Database



Python (FastAPI)
AI/ML Services



React.js
Frontend



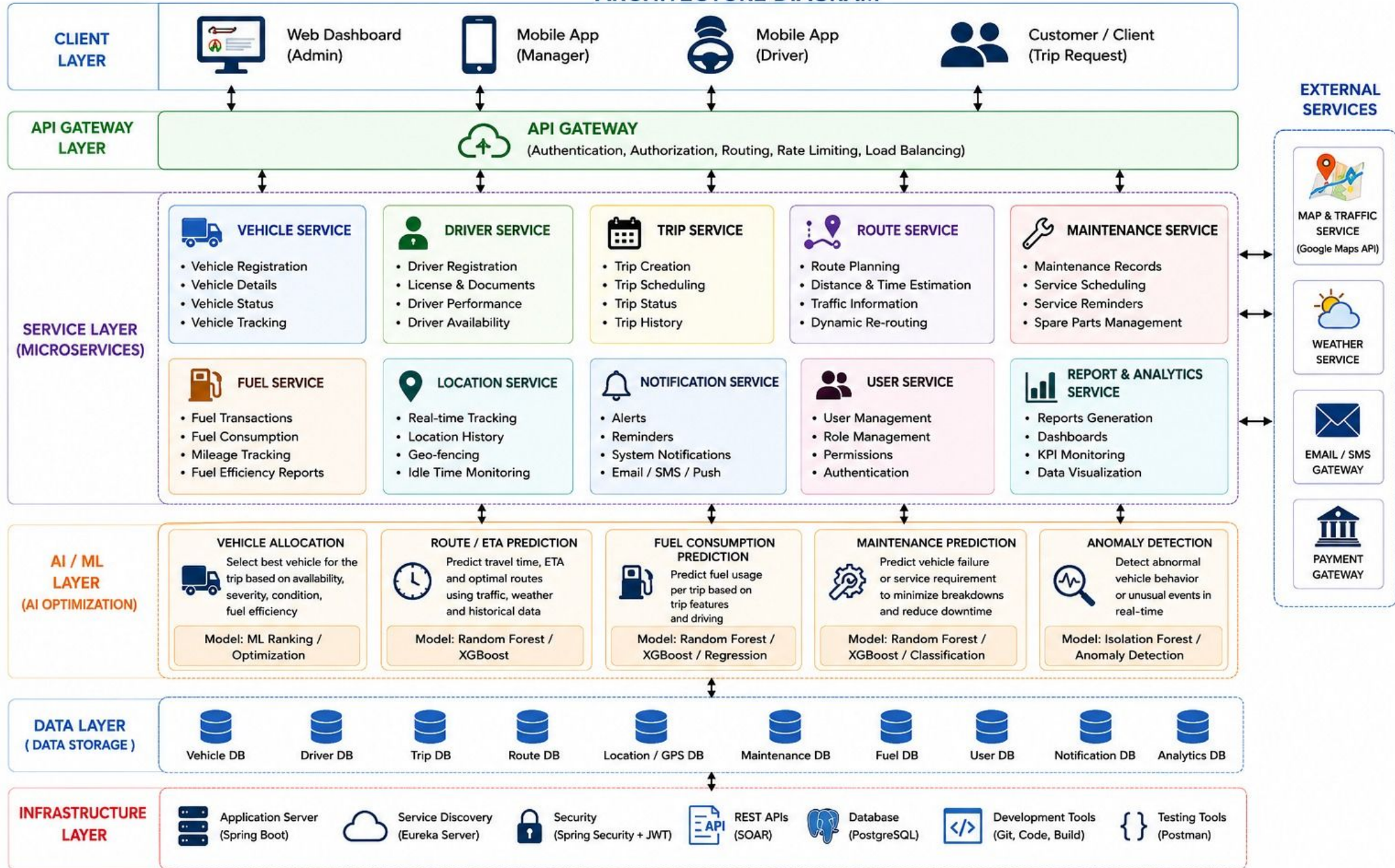
Spring Security
(JWT)



External APIs
(Maps, Weather, SMS, etc.)

AI-BASED SMART TRANSPORT FLEET MANAGEMENT SYSTEM

ARCHITECTURE DIAGRAM



Thank You

For Your Time
and Attention!



Towards Smarter Routes,
Better Journeys.



AI-Powered
Intelligence



Optimal
Routes



Secure
System



Scalable
Architecture

